

patterning with immersion lithography has already been used where EUV had previously been expected to be used. However, it is currently recognised that EUV cannot practically realise 40nm to 50nm pitch due to stochastic effects in resist exposure, so even 10nm node is off limits. End of the road for silicon is nearing as alternative materials are now required for 7nm node and beyond.

Use of EUV lithography in a manufacturing plant today is almost as innovative as making 7nm SiGe based transistors. These use light in the DUV range at about 248nm wavelength to print 150nm to 120nm-size features on a chip. Beyond that point,

smaller features require wavelengths in EUV range.

Light at 193nm and 157nm wavelengths is absorbed instead of transmitted by conventional lenses. The result is no light, no image and

no circuit. To reach these small geometries, manufacturers use self-aligned quadruple patterning and EUV lithography. EUV lithography is a next-generation lithography technology using EUV wavelength, currently expected to be 13.5nm.

Current lithographic techniques utilise DUV light sources that produce wavelengths of 248nm or 193nm. EUV light used in EUV lithography has a wavelength of 13.5nm, a full order of magnitude shorter.

This is important because imaging capabilities or resolutions of a lithography system are proportional to (and limited by) the wavelength of light used.

*To be continued...*

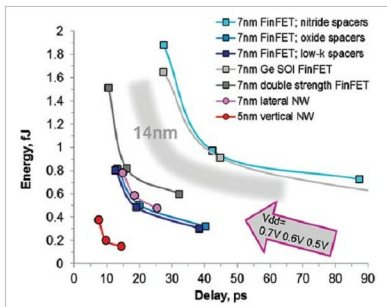


Fig. 4: Energy versus delay

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